

Hemanthkumar Vinnakota

Data Engineer

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PROFESSIONAL SUMMARY

Data Analyst with 4+ years of experience transforming raw data into actionable insights that drive business growth and efficiency. Proficient in SQL, Python, and R, with a strong command of data visualization tools like Tableau and Power BI. Skilled in statistical analysis, A/B testing, and predictive modeling to uncover trends and optimize performance. Adept at automating reports, streamlining workflows, and building interactive dashboards to enhance decision-making. Proven track record of collaborating with stakeholders to solve complex problems and translate data into strategic opportunities. Passionate about leveraging data to create measurable impact and support data-driven business decisions.

TECHNICAL SKILLS

Programming & Scripting: Python, Java, SQL, R, NoSQL, DAX, PowerShell, Bash

Data Analysis & Statistics: A/B Testing, Hypothesis Testing, Regression Analysis, Predictive Modeling, Time Series Analysis, Bayesian Statistics, Statistical Monte Carlo Simulation, Sentiment Analysis

Data Manipulation: NumPy, Pandas, Matplotlib, Seaborn, SciPy, Dask, Data Cleaning & Preprocessing, Data Aggregation, Anomaly Detection

Data Visualization: Power BI, Tableau, Looker, D3.js, Plotly, ggplot2, Excel (Pivot Tables, Macros, VLOOKUP)

Databases & Cloud Technologies: MS-SQL, Oracle, MySQL, PostgreSQL, MongoDB, Redshift, Snowflake, AWS, Microsoft Azure, Data Warehousing

Big Data: Apache Spark, PySpark, Hadoop, Hive, Kafka, Flink, Airflow, Databricks, Stream Processing

Data Engineering & Pipeline Development: ETL Pipeline Development, Data Modeling & Schema Design, Database Replication & Synchronization, API Integration, Real-time Data Processing

ML & AI: Supervised & Unsupervised Learning, Deep Learning, Reinforcement Learning, Feature Selection, Model Optimization, Anomaly Detection

Software Development & DevOps: Flask, Django, FastAPI, REST APIs, Docker, Kubernetes, Git, GitHub, GitLab, CI/CD Pipelines, Jenkins, Terraform, Serverless Architecture, Agile, Scrum, Jira

Soft Skills: Data Storytelling & Insight Communication, Stakeholder Management, Problem-Solving, Cross-Functional Collaboration, Revenue Forecasting, Risk Analysis, Operations Optimization

EXPERIENCE

Data Engineer/Data Analyst, DXC Technology | Remote, USA

Aug 2023 – Present

- Optimized customer analytics models by integrating machine learning techniques, enhancing predictive accuracy by 25%, which enabled more effective targeted marketing campaigns and increased customer retention rates.
- Developed and maintained 20+ interactive dashboards in Power BI and Tableau, automating real-time reporting and reducing manual reporting efforts by 40 hours per month, leading to faster decision-making.
- Designed and implemented scalable ETL pipelines to automate data ingestion from 5+ disparate data sources, reducing data processing time by 60% and improving system reliability.
- Led company-wide A/B testing initiatives on promotional strategies, analyzing customer response data that resulted in a 15% increase in revenue per customer and improved sales forecasting accuracy.
- Implemented an advanced anomaly detection algorithm in SQL and Python, reducing data discrepancies by 30%, ensuring clean and accurate datasets for high-stakes decision-making.
- Developed ad-hoc and scheduled SQL queries to analyze healthcare claims data, reducing fraudulent claims by 22% through enhanced anomaly detection techniques.
- Designed and optimized data models for efficient storage and retrieval, improving query performance by 35% for large-scale customer analytics datasets.
- Implemented CI/CD pipelines for ETL workflows using Apache Airflow and GitHub Actions, reducing deployment time for data pipeline updates by 50%.
- Developed scalable batch and streaming data pipelines using Apache Spark and Kafka, enabling near real-time analytics for customer insights and fraud detection.
- Engineered data warehouse solutions in Snowflake, optimizing storage costs and query performance, leading to a 20% reduction in cloud expenses.
- Integrated raw transactional data into a centralized data lake using AWS Glue, facilitating seamless access to structured and semi-structured data for analysts and data scientists.

Junior Data Analyst, Infosys BPM | India

Oct 2019 – Dec 2021

- Extracted, transformed, and analyzed customer behavior data from 2M+ records, identifying key purchasing trends that influenced marketing strategies, resulting in an 18% increase in customer engagement.

- Automated monthly and quarterly business intelligence reports using SQL and Python scripts, reducing manual reporting time by 50%, allowing executives to focus on strategic decision-making.
- Developed predictive machine learning models for sales forecasting, improving demand accuracy by 20%, which led to a more efficient supply chain and reduced excess inventory costs.
- Designed, built, and maintained over 10+ dynamic dashboards in Power BI to track KPIs, sales performance, and operational metrics, enabling business leaders to make data-driven decisions 35% faster.
- Enhanced data quality by implementing automated data cleaning and validation processes, reducing data inconsistencies by 95%, which improved model reliability and business intelligence accuracy.
- Performed root cause analysis on customer churn data, identifying critical factors contributing to attrition, and proposed retention strategies that led to a 10% decrease in churn rates.
- Conducted cohort analysis and customer segmentation using clustering techniques, providing insights that allowed the marketing team to personalize campaigns, leading to a 25% increase in conversion rates.
- Developed ELT processes using dbt (Data Build Tool), ensuring efficient transformation and modeling of marketing and sales data for BI reporting.
- Optimized SQL queries and indexing strategies, reducing query execution time by 60% in PostgreSQL-based analytics databases.
- Automated data pipeline monitoring and alerting using Prometheus and Grafana, reducing downtime by 40% and improving pipeline reliability.
- Built custom API integrations to ingest third-party marketing and CRM data into internal databases, enabling cross-platform analytics and reporting.

EDUCATION

Masters in Computers and information sciences

Jan 2022 – May 2023

Southern Arkansas University | Magnolia, AR, USA

Bachelor of Engineering in Computer Science

Aug 2016 – May 2020

Lovely Professional University | Punjab, India

PROJECTS

Estimated Traffic through satellite images using Detectron-2 and R-CNN Model

Utilizing high-resolution satellite imagery, a thorough traffic estimation model was established, demonstrating the importance of reliable data in obtaining successful outcomes.

Skin Lesion classification using CNN and Transfer Learning

Developed and implemented a deep learning-based skin cancer classification project with 97% accuracy using a custom 17-layer CNN and transfer learning with VGG19, EfficientNetB7, and ResNet152V2 on the HAM10000 dataset.

Prediction of Bubble Point Pressure and Oil Formation Volume Factor Using Machine Learning

The project included a thorough Exploratory Data Analysis (EDA) phase in which I investigated the properties and connections of the dataset. I improved prediction accuracy by highlighting important patterns and connections through smart visuals that helped with feature engineering and model selection.